Dataset Analysis

Introduction

This analysis of the Titanic Dataset and the Students Performance Dataset to understand the data structure, types, and ML readiness of datasets.

Methodology

1. head() → Displays the first few rows of the dataset
   tail() → Displays the last few rows

2. Titanic dataset- Numerical- Age, Fare, SibSp, Parch-
   Categorical- Sex, Name- Ordinal- Pclass- Binary- Survive.
   Student Performance dataset- Numerical- Math, Reading, Writing Scores- Categorical - Gender, Race, Lunch- Ordinal- Parental Education- Binary- Test Prep.

3. df.info()- was used to inspect data types and missing values
   df.describe()- provided statistical summaries.

4. Identifies number of categories and Detects imbalanced classes

5. Titanic- Target Survived and Input features- Age, sex, fare, pclass
   Student- Target Math/Reading/Writing Score and Input features- Gender, Education

6. Titanic
   - Dataset size- Rows: ~800–900 passengers, Columns: ~10–12 features
   - Medium-sized dataset, suitable for binary classification and exploratory data analysis (EDA), logistic regression
   - Students Performance dataset
   - Dataset Size- Rows: ~1,000 student records, Columns: 8 features
   - Well-structured dataset, Balanced number of samples
   - Suitable for regression and classification, Linear Regression

7. Titanic Dataset
   - Age and class imbalance, name has high cardinality, fare contains extreme values
   - Students Performance Dataset
   - No missing values, multiple categorical columns require encoding

Conclusion

Both datasets are suitable for ML algorithms like Logistic Regression and Linear Regression.